Grid-Aware Smart EV Charging

Scheduling charging currents with convex optimization

How can a workplace charge its EVs without creating an expensive demand peak?

Tarun Thomas Tom · Milan V Raj · Nirmal Thomas Mathew

Real charging data / CVXPY + Clarabel / Five-minute control

Why charging on arrival is a problem

Many arrivals, one shared connection

Most drivers arrive between 7 and 10 a.m. Charging immediately concentrates the load.

The bill depends on the peak

The tariff adds $15.51 per kW of the month’s highest 15-minute average demand.

Parking time gives us flexibility

Vehicles often stay longer than the time needed to charge. We can spread the demand.

Uncontrolled September bill



The demand charge is larger than the energy charge.

RELATED WORK

What we build on

Adaptive Charging Networks

Lee et al. (2021): convex MPC with three-phase network limits. This is our starting framework.

Valley-filling charging

Gan, Topcu and Low (2013): flatten aggregate demand with a convex load objective.

Data and simulation tools

ACN-Data supplies sessions; ACN-Sim evaluates charging schedules.

What this project studies

1. Monthly peak-demand billing
2. Uncertain driver requirements
3. Solar generation and forecast errors

System and experimental setup

Item	Setting
Charging facility	Caltech ACN garage
Network	54 chargers; 3 phases; 150 kVA
Dataset	1,772 sessions; 19 weekdays
Study period	September 2018
Control / horizon	5 minutes / up to 12 hours

Three main experiments

S2 · Monthly billing

Known needs and departures of connected EVs.

S3 · Predicted drivers

Requirements estimated from the previous four weeks.

S4 · Add solar

50 kW PV; forecast output; bill on net grid import.

Decision variables and constraints

Choose R(i,t): the charging current of EV i in interval t.

Constraint	Mathematical form	Meaning
Charging limit	$0 \le R(i,t) \le \bar{r}(i,t)$	Zero outside the plug-in window
Energy requirement	$\Sigma_t E(i,t) + u(i) = e(i)$	$u(i) \ge 0$ is unmet energy
Network limits	$\lVert B(m)\, R(:,t) \rVert_2 \le c(m)$	Phase and transformer limits

Power: $P(t) = \Sigma_i V(i) R(i,t) / 1000$ | Energy: $E(i,t) = V(i) R(i,t) \Delta / 1000$

Unmet energy is allowed but penalized, so tight charging windows need not make the problem infeasible.

What does the optimizer minimize?

Electricity cost + charging shortfall + optional grid-quality penalties

BILL-LP

Time-of-use energy cost, demand charge, peak-exceedance penalty and unmet energy.

COMBINED-QP

The same terms, plus quadratic penalties for load flattening and phase imbalance.

Why the problem is convex

Linear charging constraints

Convex max and hinge costs

Convex quadratic penalties

Second-order cone network limits

Despite its name, BILL-LP is an SOCP once the network constraints are included.

Report §IV-D; Table I • Full monthly-demand expression in backup slide 13

How we solve it in practice

1 Observe

Connected EVs, energy delivered, paid peak

2 Optimize

Plan charging over a horizon of up to 12 h

3 Apply

Use only the first 5-minute decision

4 Repeat

Update information and solve again

Implementation stack

Python + CVXPY

Variables, objective and constraints

Clarabel

Primary conic solver; SCS as fallback

ACN-Sim + ACN-Data

Simulation and historical sessions

MPC is the control strategy, not the solver.

Monthly billing: lower peak, lower bill

Controller	Bill	15-min peak	Delivery
Round Robin (RR)	$4,088	142.5 kW	99.8%
MPC-BILL-LP	**$3,031**	**70.3 kW**	**98.9%**
MPC-COMBINED-QP	$3,518	77.6 kW	96.5%
Offline OPT-BILL-LP	$3,150	71.2 kW	99.6%

Compared with RR: $1,057 lower bill (26%) and 51% lower billed peak.

Trade-off: delivery falls by 0.9 percentage points. Read savings together with service.

Predictions alone can leave EVs undercharged

MPC-COMBINED-QP, with different models of driver behavior

Information model	Bill	Delivery
Median prediction	$2,782	76.4%
Conservative prediction	$3,470	94.3%
Historical scenarios (SAA-30)	**$3,178**	**96.3%**

SAA penalizes expected unmet energy across up to 20 historical scenarios per EV.

Delivery rises by 19.9 percentage points over Median. A low bill alone is not success.

Solar helps, but forecast errors can raise the peak

Why the peak can keep rising

1. Schedule close to the peak already paid for.

2. Solar output falls below forecast.

3. Grid import sets a higher peak for the rest of the month.

Flattening and scenario-based scheduling reduce this effect in the tests.

Selected online configurations

Controller / info	Bill	Delivery
Bill / Perfect	$2,514	98.6%
Combined / SAA+F	**$1,904**	**97.8%**

Billed import peak: 97.2 → 60.2 kW

SAA+F uses scenarios and a corrected future-arrival forecast.

COMPUTATION

Can it finish before the next control step?

The controller has 300 seconds between charging updates.

Online configuration	Median	99th percentile	Maximum
S2 · BILL-LP	0.09 s	2.29 s	3.8 s
S3 · SAA-30, Combined	0.15 s	8.27 s	10.3 s
S4 · SAA-30, Combined+F	1.31 s	12.79 s	21.4 s

All tested online solves fit within the five-minute interval.

The offline joint-month benchmark takes about 10–17 minutes and is not an online controller.

Report Table II • Wall time includes model build, CVXPY compilation and solve

CONCLUSION

What we learned—and what remains to test

Useful outcomes

Cost and peak reduction

About 26% lower bill and 51% lower billed peak than the cheapest heuristic in S2.

Better delivery with uncertainty

Historical scenarios recover much of the delivery lost with median predictions.

Limitations

One month at one site

Synthetic phase assignment and PV weather

Continuous, not discrete, charging currents

Scenario weight tuned on the test month

No PV-aware heuristic baseline

Next: test other months, tune weights out of sample and handle real charger pilot levels.

Monthly demand and scenario shortfalls

15-minute billed demand and the planning floor

$$\bar{P}(t) = [P(t) + P(t-1) + P(t-2)] / 3$$

$$F = \max\{P^*, F_{mtd}\} \quad P^* = 54.93 \text{ kW}$$

Demand epigraph: $q \geq \bar{P}(t)$, $q \geq F$ Add $d \cdot q$ to the objective.

Scenario-based expected unmet energy

$$z(i,s) \geq k(i,s) - C(i,d(i,s)), \quad z(i,s) \geq 0$$

Penalty: $w \, \Sigma_i \, \Sigma_s \, \pi(i,s) \, z(i,s)$ with $w = 30$ \$/kWh

With PV, billing uses net import $S(t) \geq P(t) - PV_{forecast}(t)$, $S(t) \geq 0$.

Additional results for comparison

S2 · Monthly billing, known connected-driver requirements

Controller	Bill	Peak	Delivery
Uncontrolled — infeasible	$4,845	191.6 kW	100.1%
Earliest Deadline First	$4,116	144.0 kW	99.9%
Least Laxity First	$4,096	142.9 kW	99.9%

S4 · Solar cases illustrating the cost / delivery trade-off

Controller / information	Bill	Peak	Delivery
Combined-QP / Median	$1,629	54.9 kW	75.8%
OPT-BILL-LP / Perfect	$1,675	55.1 kW	99.6%

References

1. Lee et al. (2021). Adaptive charging networks: A framework for smart electric vehicle charging. IEEE Transactions on Smart Grid, 12(5), 4339–4350.

2. Lee, Li & Low (2019). ACN-Data: Analysis and applications of an open EV charging dataset. ACM e-Energy, 139–149.

3. Lee et al. (2021). ACN-Sim: An open-source simulator for data-driven electric vehicle charging research. IEEE Transactions on Smart Grid, 12(6), 5113–5123.

4. Gan, Topcu & Low (2013). Optimal decentralized protocol for electric vehicle charging. IEEE Transactions on Power Systems, 28(2), 940–951.

5. Diamond & Boyd (2016). CVXPY: A Python-embedded modeling language for convex optimization. Journal of Machine Learning Research, 17(83), 1–5.

6. Goulart & Chen (2024). Clarabel: An interior-point solver for conic programs with quadratic objectives. arXiv:2405.12762.